



How mars lost water

Towers of dust 50 miles high which act like space elevators might explain how water escaped Mars. <https://digit.in/dec19-55>



Cybertrucks in the Police

Dubai is adding's Musk's newest Tesla model, the Cybertruck, to their police fleet come 2020. <https://digit.in/dec19-56>



BIOINSPIRED SPACECRAFT

Concepts derived from 3 billion years of evolution

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ecause of the ruthless forces of natural selection, organisms have developed all kinds

of strategies to interact with their environments. Outer space is not conducive to life, but aerospace engineers can reverse engineer applicable biological approaches in robots, and then apply them to space exploration. Robotic probes and living things are often constrained in similar ways, such as a requirement for optimum energy use, a tolerance for extreme environments, or the ability to make rapid decisions in novel situations. Almost every biological system that has evolved has some kind of application in space. Sense organs, bones and appendages for locomotion in animals can serve as inspiration for sensors, structural components and propulsion systems in spacecraft. One of the most famous people to use this biomimetic approach for designing new machines, was Leonardo Da Vinci. Note

that the title images are in the style of Da Vinci, but not his actual sketches.

ROVERS

Space agencies around the world, including NASA and ESA are collaborating with research institutions to develop spacecraft concepts that can turn into planned missions in the future. NASA and Cornell University are developing soft robots to explore the global subsurface oceans in the moons of the gas giants. The Jovian moon of Europa is the prime target for such missions. The concepts being developed derive their energy from the environment, one possibility being the strong magnetic field

of Jupiter, harvested through the use of electrodynamic tentacles that serve as generators. Such an aquatic rover could derive its fuel from the ambient water, which is available in abundance. After electrolyzing the water, the hydrogen and oxygen can be stored in the body and limbs of the rover. Sparks could be used to make these gasses expand, allowing the rover to propel itself. The approach would mean that the rover would not require to have any solar or nuclear power system on board, allowing the robot to stay functional for longer periods of time. For lighting, the researchers are using a stretchable electroluminescent film, similar to bioluminescence in marine animals, allowing the imaging sensors on board to record the local environment. The approaches of power generation and locomotion by expanding gas are both breakthrough concepts, but are feasible within the limits of current technologies. Researchers at Cornell are considering two options,



The concept of a lamprey robot for underwater exploration of Europa

Credit: Cornell University